IBM PROJECT REPORT

Telecom Churn Driver Discovery & Persona Profiler

Name	Varun Bale
Roll No	A24126552249
IBM ID	IBMQ2DST1667
Department	CSE (AI & ML)
College Name	Anil Neerukonda Institute of Technology and Sciences
Module Name	Artificial Intelligence and Machine Learning
Module Number	16
UG Level	UG2

Table of Contents

1. Executive Summary

Customer churn is one of the most pressing challenges faced by telecommunication providers, as acquiring a new subscriber is significantly more expensive than retaining an existing one. This project, undertaken as part of the IBM Case Study module on Artificial Intelligence and Machine Learning, presents “TeleChurn AI” — an end-to-end Telecom Churn Driver Discovery and Persona Profiler platform. The system analyses the Telco Customer Churn dataset to identify the structural business rules behind customer cancellations using an interpretable Decision Tree classifier, and further segments the subscriber base into meaningful behavioural personas using K-Means clustering. The resulting web application provides telecom operators with a live dashboard, a searchable customer registry, churn-driver analytics, an interactive churn-risk predictor, and exportable operational reports — enabling data-driven retention strategies rather than reactive customer-service responses.

Across the analysed base of 7,043 customers, the platform surfaces an overall churn rate of 26.54%, identifies month-to-month contracts, low tenure, high monthly charges and the absence of technical support as the dominant churn drivers, and classifies customers into four behavioural personas ranging from highly loyal, low-risk subscribers to a New/At-Risk segment churning at over 68%. The trained CART decision tree achieves 81.3% training accuracy and 78.9% test accuracy with an F1-score of 0.74 for the churn class, providing both predictive power and full rule-level interpretability for business stakeholders.

2. Introduction & Problem Statement

Telecom providers operate in a highly saturated and competitive market where subscriber attrition directly erodes recurring revenue. Traditional churn-analysis approaches rely on “black-box” models that predict who will churn but fail to explain why, leaving retention teams without actionable levers to pull. Business teams need clear, human-readable rules — for example, “customers on month-to-month contracts with less than 12 months of tenure and no technical support are at very high risk” — that can be translated directly into retention offers, bundling strategies, and proactive outreach campaigns.

The objective of this case study is therefore twofold: first, to train an interpretable decision-tree model that exposes the concrete business rules driving cancellations; and second, to group the existing customer base into distinct behavioural personas so that marketing and retention teams can design segment-specific interventions rather than a one-size-fits-all approach.

3. Objectives

- **Driver Discovery:** Train a CART decision-tree classifier on the Telco Customer Churn dataset to uncover the key, interpretable rules that lead to customer cancellation.
- **Persona Profiling:** Apply K-Means clustering on behavioural and billing attributes to segment customers into distinct personas (e.g., Loyal, Premium, Price-Sensitive, New/At-Risk).
- **Risk Prediction:** Provide an interactive churn-risk assessment tool that estimates the probability of churn for a given customer profile in real time.

- **Actionable Reporting:** Present all findings through a live analytics dashboard and generate downloadable reports/CSV exports for business consumption.
- **Model Transparency:** Ensure every prediction can be traced back to a human-readable decision path rather than an opaque probability score.

4. Dataset Description

The project uses the publicly available Telco Customer Churn dataset, which contains 7,043 customer records describing account, service and billing attributes for a fictional telecommunications company. Each record represents one subscriber and is labelled with whether that customer churned within the observed period.

7,043 Total Customers	**1,869** Churned Customers	**26.54%** Overall Churn Rate	**4,036** High-Risk Customers

Key Feature Groups

- **Demographic attributes:** gender, senior citizen status, partner and dependents.
- **Account attributes:** tenure (months), contract type (month-to-month, one year, two year), paperless billing and payment method.
- **Service attributes:** internet service type (DSL, Fiber optic, None), online security, tech support, streaming TV/movies, and related add-ons.
- **Billing attributes:** monthly charges and total charges.
- **Target label:** Churn (Yes/No), used to train and evaluate the classifier.

5. System Architecture & Technology Stack

TeleChurn AI is delivered as a secure, role-based web application. Users authenticate via an email/password login screen before accessing the analytics workspace. The backend serves the trained models and pre-computed analytics, while the frontend renders an interactive dashboard, data tables, charts and forms consistent with a modern SaaS-style admin console.

Application Modules

- **Dashboard —** high-level KPIs and churn-by-contract / churn-by-tenure summaries.
- **Customers —** a searchable, filterable registry of all subscribers with risk and persona tags.
- **Churn Analytics —** driver analysis by contract, payment method, internet service and tenure band.
- **Churn Drivers —** the extracted decision-tree business rules and risk-factor impact ranking.
- **Personas —** K-Means behavioural segments with churn rate, tenure and spend profiles.
- **Churn Prediction —** an interactive form that scores a hypothetical customer's churn risk in real time.
- **Model Performance —** the decision-tree structure, accuracy metrics and decision-path visualisation.
- **Reports —** operational summaries with a downloadable CSV export.

- **Profile —** account and role management for platform users.

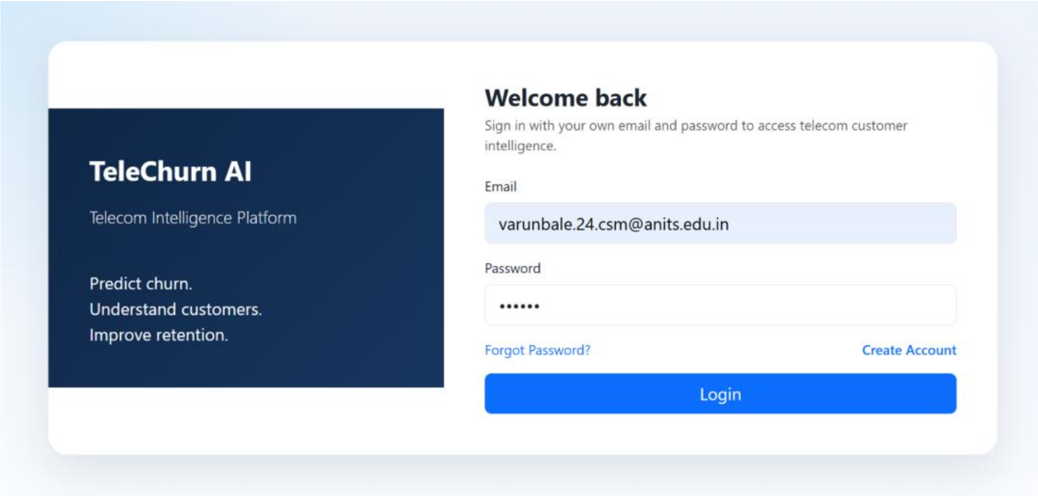


Fig. 5.1 — Secure login screen of the TeleChurn AI platform

6. Methodology

6.1 Data Preprocessing

Raw customer records were cleaned to handle missing/blank TotalCharges values, categorical fields were standardised, and derived features such as tenure bands (0–12, 12–24, 24–48, 48+ months) were engineered to support both the decision-tree model and downstream analytics visualisations.

6.2 Decision Tree Model (CART)

A Classification and Regression Tree (CART) classifier was trained with Gini impurity as the splitting criterion and a maximum depth of 4, chosen deliberately to preserve interpretability over marginal accuracy gains. Each split in the resulting tree corresponds to a concrete, explainable business condition (e.g., contract type, tenure, monthly charges, tech support), and each leaf reports the churn probability and the number of customers following that path.

6.3 K-Means Persona Clustering

Customers were grouped into behavioural personas using K-Means clustering over tenure, monthly charges and service-usage features. The resulting clusters were manually labelled based on their statistical profile — Loyal Customers, Premium Customers, Price-Sensitive Customers, and New/At-Risk Customers — giving business teams an intuitive vocabulary for retention planning.

6.4 Churn Risk Prediction

The trained decision tree is exposed through an interactive “Quick Churn Risk Assessment” form, where a user enters tenure, contract type, monthly charges, internet service and tech-support status to receive an instant risk score, a plain-language explanation of the contributing factors, and a recommended retention action.

7. Application Walkthrough

This section presents the core screens of the TeleChurn AI platform, illustrating how the underlying models are surfaced to end users.

7.1 Dashboard

The Dashboard provides an at-a-glance overview of customer intelligence: total customers, churned customers, overall churn rate, and the count of high-risk customers, alongside churn-by-contract and churn-by-tenure-band breakdowns and a ranked list of the top churn drivers.

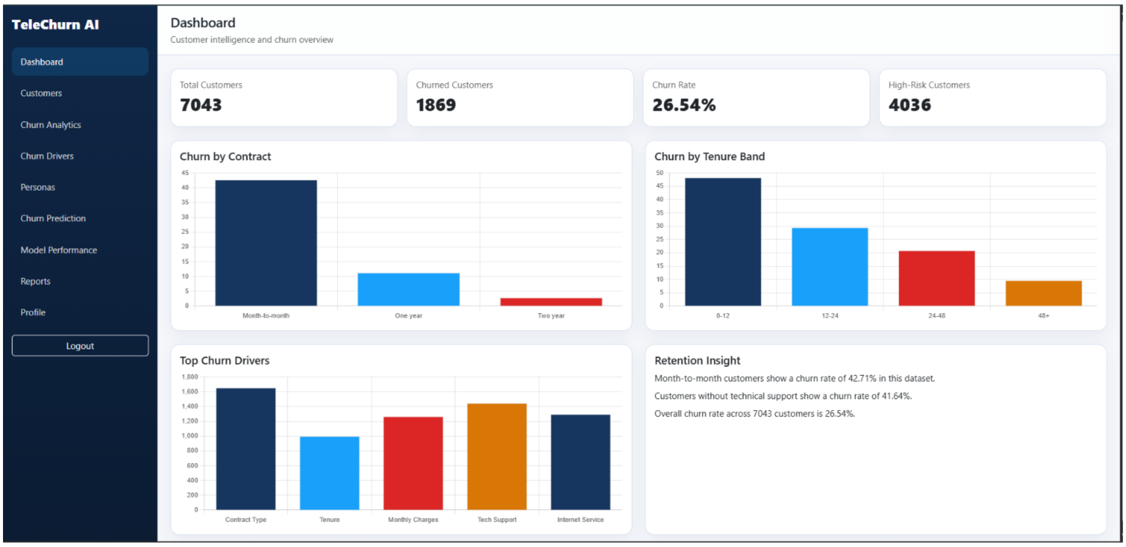
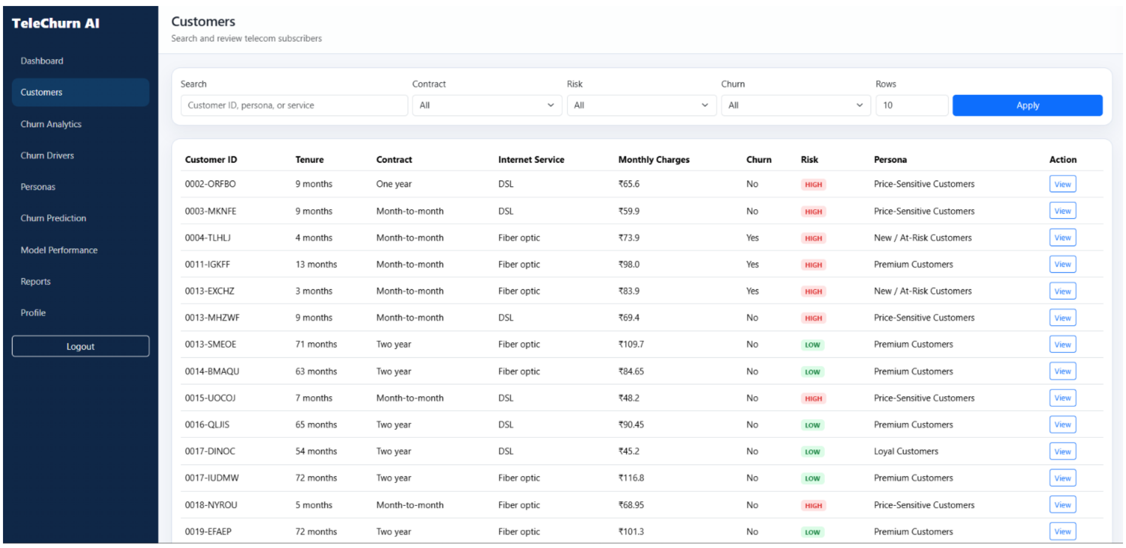


Fig. 7.1 — Dashboard: customer intelligence and churn overview

7.2 Customers

The Customers module offers a searchable, filterable registry of all subscribers, supporting filters by contract type, predicted risk level and churn status, with each customer tagged by their assigned persona for quick triage.



Customer ID	Tenure	Contract	Internet Service	Monthly Charges	Churn	Risk	Persona	Action
0002-ORFBO	9 months	One year	DSL	₹65.6	No	HIGH	Price-Sensitive Customers	View
0003-MKNFE	9 months	Month-to-month	DSL	₹59.9	No	HIGH	Price-Sensitive Customers	View
0004-TLHLJ	4 months	Month-to-month	Fiber optic	₹73.9	Yes	HIGH	New / At-Risk Customers	View
0011-IGKFF	13 months	Month-to-month	Fiber optic	₹98.0	Yes	HIGH	Premium Customers	View
0013-EXCHZ	3 months	Month-to-month	Fiber optic	₹83.9	Yes	HIGH	New / At-Risk Customers	View
0013-MHZWF	9 months	Month-to-month	DSL	₹69.4	No	HIGH	Price-Sensitive Customers	View
0013-SMEOE	71 months	Two year	Fiber optic	₹109.7	No	LOW	Premium Customers	View
0014-BMAQU	63 months	Two year	Fiber optic	₹84.65	No	LOW	Premium Customers	View
0015-UOCOJ	7 months	Month-to-month	DSL	₹48.2	No	HIGH	Price-Sensitive Customers	View
0016-QLJIS	65 months	Two year	DSL	₹90.45	No	LOW	Premium Customers	View
0017-DINOC	54 months	Two year	DSL	₹45.2	No	LOW	Loyal Customers	View
0017-IUDMW	72 months	Two year	Fiber optic	₹116.8	No	LOW	Premium Customers	View
0018-NYROU	5 months	Month-to-month	Fiber optic	₹68.95	No	HIGH	Price-Sensitive Customers	View
0019-EFAEP	72 months	Two year	Fiber optic	₹101.3	No	LOW	Premium Customers	View

Fig. 7.2 — Customers: search and review telecom subscribers

7.3 Churn Analytics

The Churn Analytics screen breaks down churn by contract type, internet service, payment method and tenure band, surfacing the business insight that month-to-month customers churn at 42.71% and customers without technical support churn at 41.64%.

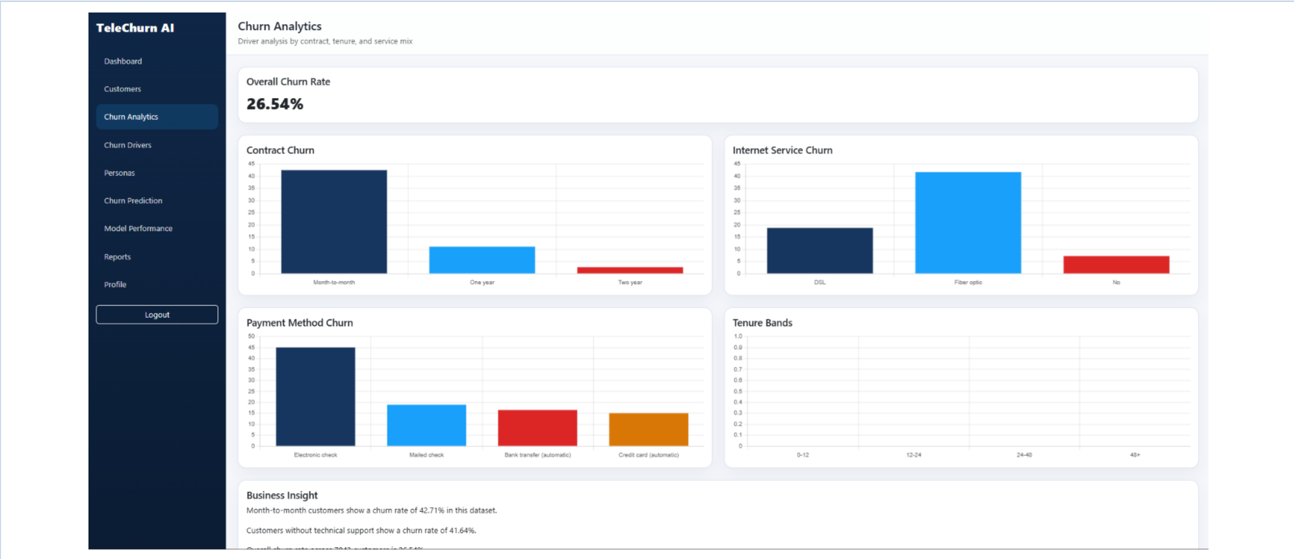


Fig. 7.3 — Churn Analytics: driver analysis by contract, tenure and service mix

7.4 Churn Drivers

The Churn Drivers page exposes the extracted decision-tree rules directly to the business user — ranking risk factors by impact and translating each rule into a recommended action for high-risk and all-customer segments alike.

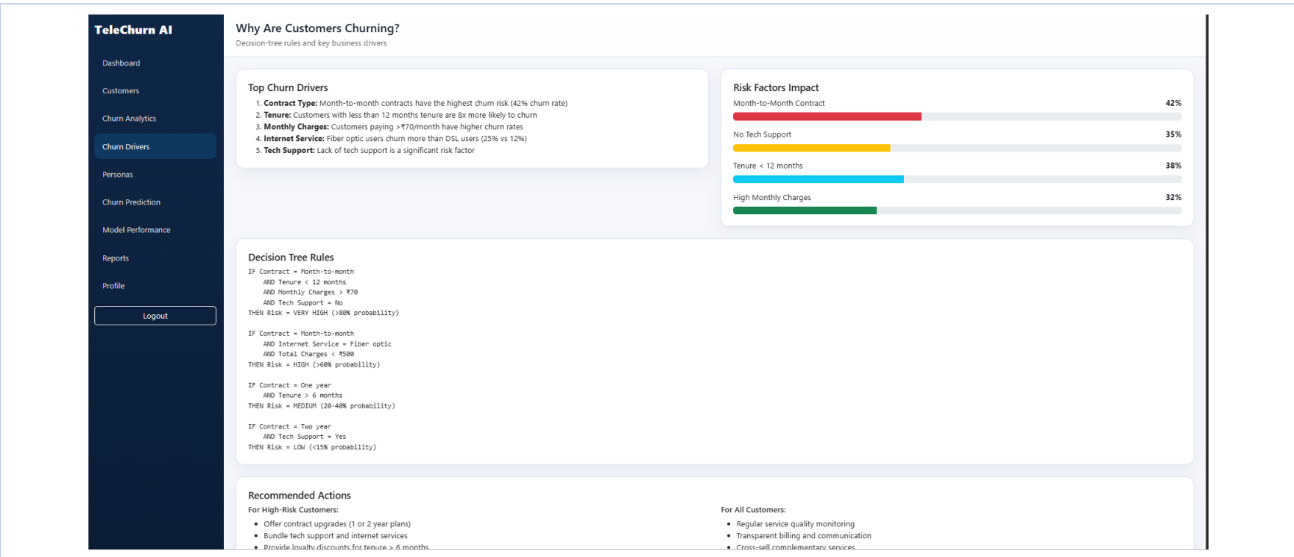


Fig. 7.4 — Churn Drivers: decision-tree rules and key business drivers

7.5 Churn Prediction

The interactive risk-assessment form allows a user to input a hypothetical customer's tenure, contract, monthly charges, internet service and tech-support status, returning an instant churn-risk percentage, contributing factors and a recommended retention action.

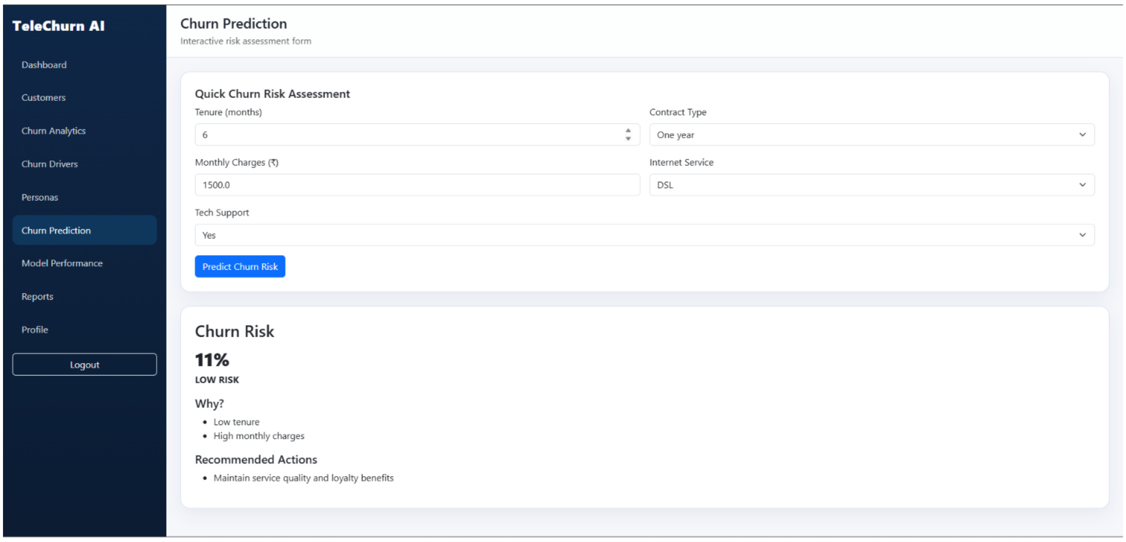


Fig. 7.5 — Churn Prediction: interactive risk assessment form

7.6 Model Performance

The Model Performance page visualises the trained decision tree itself — an expandable node diagram — alongside training accuracy, test accuracy and the F1-score for the churn class, giving full transparency into how each prediction is derived.

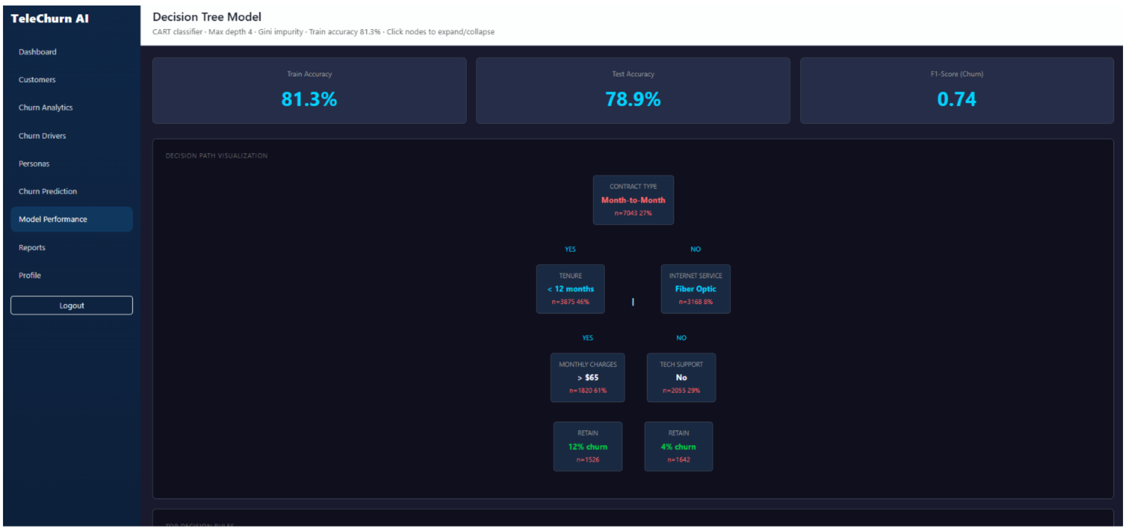


Fig. 7.6 — Decision Tree Model: CART classifier performance and decision path

7.7 Personas

The Personas screen presents the four K-Means behavioural segments — Loyal, New/At-Risk, Premium and Price-Sensitive Customers — each annotated with customer count, churn rate, average tenure, average monthly charges and a recommended retention strategy.

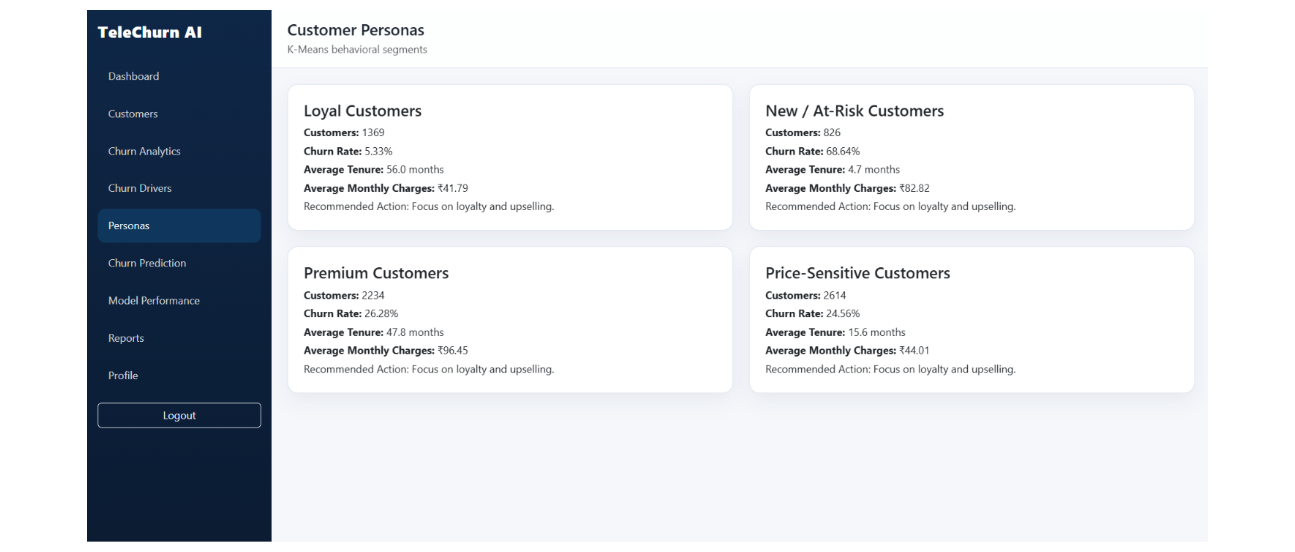


Fig. 7.7 — Customer Personas: K-Means behavioural segments

7.8 Reports

The Reports module consolidates operational summaries and provides a downloadable CSV export of the full customer-level dataset — including tenure, contract, service, billing, churn, risk and persona fields — for offline analysis or integration with CRM systems.

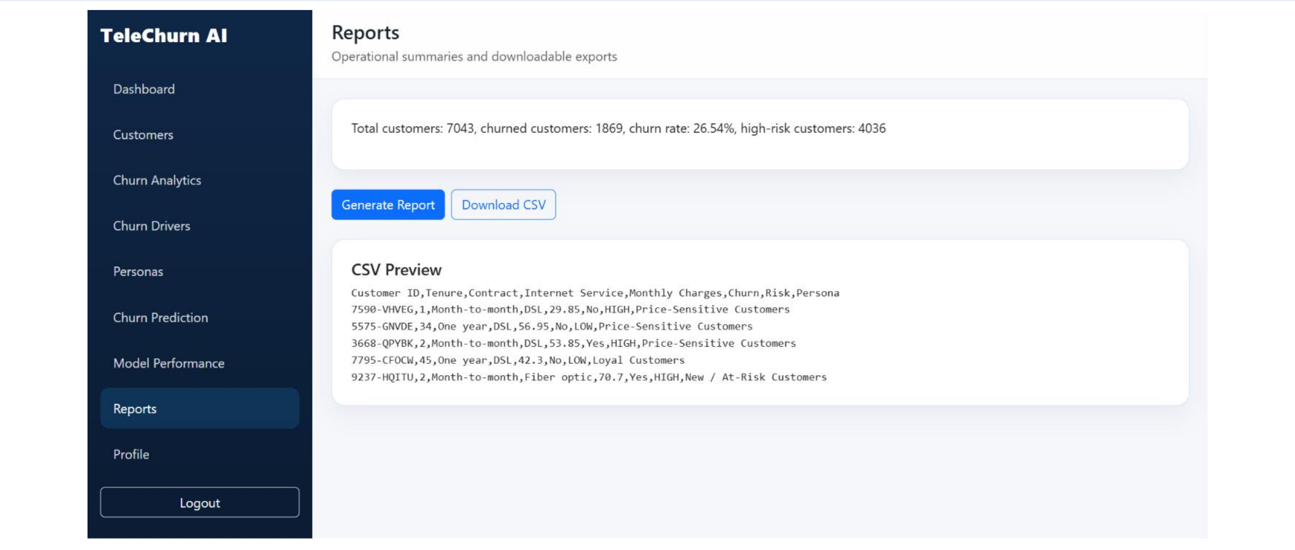


Fig. 7.8 — Reports: operational summaries and downloadable exports

7.9 Profile

The Profile module allows an authenticated user to manage their account details, update their password, and view their assigned platform role.

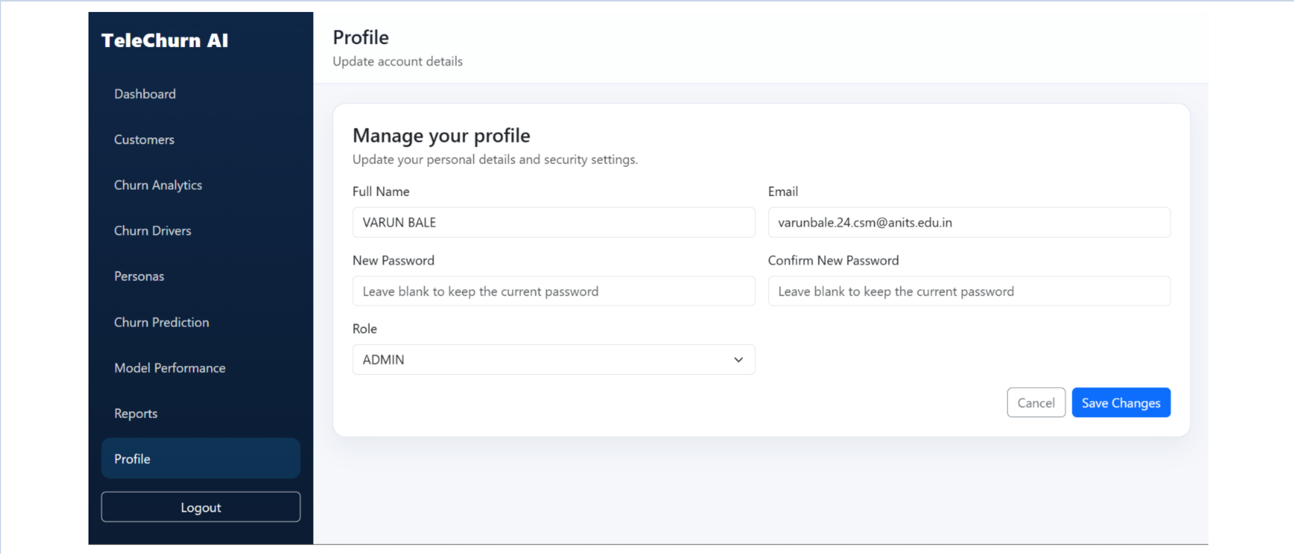


Fig. 7.9 — Profile: update account details

8. Results & Model Performance

The CART decision-tree classifier, trained to a maximum depth of 4 using Gini impurity, achieved strong and well-generalised performance on the Telco Customer Churn dataset:

81.3% Train Accuracy	**78.9%** Test Accuracy	**0.74** F1-Score (Churn)	**Depth 4** Tree Complexity

Top Decision-Tree Rules

- IF Contract = Month-to-month AND Tenure < 12 months AND Monthly Charges > ₹70 AND Tech Support = No → Risk = VERY HIGH (>80% probability).
- IF Contract = Month-to-month AND Internet Service = Fiber optic AND Total Charges < ₹500 → Risk = HIGH (>60% probability).
- IF Contract = One year AND Tenure > 6 months → Risk = MEDIUM (20–40% probability).
- IF Contract = Two year AND Tech Support = Yes → Risk = LOW (<15% probability).

Risk Factor Impact

Risk Factor	Observed Churn Impact
Month-to-Month Contract	42%
Tenure < 12 months	38%
No Tech Support	35%

Risk Factor	Observed Churn Impact
High Monthly Charges (>₹70)	32%

Customer Persona Summary

Persona	Customers	Churn Rate	Avg. Tenure	Avg. Monthly Charges
Loyal Customers	1,369	5.33%	56.0 months	₹41.79
New / At-Risk Customers	826	68.64%	4.7 months	₹82.82
Premium Customers	2,234	26.28%	47.8 months	₹96.45
Price-Sensitive Customers	2,614	24.56%	15.6 months	₹44.01

The New/At-Risk persona stands out sharply, churning at 68.64% — more than 2.5 times the overall base rate — despite representing only 826 customers, making it the single highest-priority segment for immediate retention intervention. In contrast, Loyal Customers churn at just 5.33% with an average tenure of 56 months, confirming that tenure and contract commitment are the platform's strongest protective factors against churn.

9. Business Insights & Recommendations

For High-Risk Customers

- Offer contract upgrades (migrate month-to-month subscribers to 1- or 2-year plans with incentives).
- Bundle technical support and internet services at a discount to reduce the “no tech support” risk factor.
- Provide targeted loyalty discounts once a customer crosses the 6-month tenure mark, before the high-risk window closes.

For All Customers

- Maintain regular service-quality monitoring, especially for fiber-optic subscribers who show elevated churn.
- Ensure transparent billing and proactive communication to reduce price-driven cancellations.
- Cross-sell complementary services to Premium and Price-Sensitive personas to increase switching costs.

10. Conclusion

This case study demonstrates that an interpretable decision-tree model, combined with unsupervised persona segmentation, can give telecom operators far more actionable insight than a standalone black-box churn score. By exposing explicit business rules — contract type, tenure, monthly charges and tech-support status — and grouping customers into intuitive personas, the TeleChurn AI platform bridges the gap between data science outputs and the retention decisions that business teams must actually make. With a test accuracy of 78.9% and full rule-level transparency, the system provides both predictive reliability and the explainability required for real-world adoption in a customer-retention workflow.

11. References

- IBM Telco Customer Churn Dataset — Kaggle / IBM Sample Data Sets.
- Breiman, L., Friedman, J., Olshen, R., Stone, C. — Classification and Regression Trees (CART), 1984.
- MacQueen, J. — Some Methods for Classification and Analysis of Multivariate Observations, 1967 (K-Means Clustering).
- Scikit-learn Documentation — Decision Trees and Clustering Modules.